

Ethan Chang

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EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY

Berkeley, CA | August 2022 - May 2026

Bachelor's of Art in Data Science & Minor in Computer Science

Relevant Coursework: Artificial Intelligence, Linux Systems, Probability Theory & Principles of Data Science, Discrete Mathematics & Probability Theory, Linear Algebra & Differential Equations, Data Structures & Algorithms, Structure & Interpretation of Computer Programs

RELEVANT EXPERIENCE

Berkeley Discovery | Database Assistant

February 2024 - Present

- Analyzed academic/professional listings for high-priority posts and updated the database with weekly opportunities for **6500+** concurrent users, boosting platform activity by **50%**.
- Manage **1200+** active listings and identify infrastructure bottlenecks, implementing data quality improvements and scalable UI/system updates to support growing user bases.

Innovative Design at Berkeley | Frontend Developer & Web Designer

September 2023 - May 2024

- Led graphic UX redesigns of the Berkeley Undergraduate website and designed a mobile application prototype, prioritizing intuitive and accessible user interfaces.
- Built responsive client web pages in **React.js** using **HTML** and **CSS**, integrating aesthetic high fidelity designs for dynamic content flow through strong team collaboration.

Taiwanese American Student Association | Web Officer

December 2022 - December 2023

- Troubleshooted **Linux** server issues and routinely updated relational databases through **SQL** for uptime reliability.
- Developed interactive features and content using **HTML**, **CSS**, and **JavaScript**, improving overall user engagement.
- Monitored website traffic and user analytics to drive scalability and functionality to match the organization's focus.

PROJECTS

Neural Network Modeling

November 2024

- Engineered binary perceptron, language identification, and convolutional neural network models using **PyTorch**, **numPy**, and **Matplotlib** through **Python** to solve prediction and classification tasks.
- Constructed a regression model to approximate a Sine wave using a ReLU activation function and a MNIST handwritten digit classification model, achieving a **98%** model accuracy.

Outreach Effectiveness Data Analysis | Berkeley Discovery

July 2024 - August 2024

- Leveraged **Pandas** data functions through JupyterLab, evaluating database user login activity to inform data-driven decisions about event funding allocation and effort.
- Conducted exploratory data analysis (**EDA**) to identify key patterns between time periods, visualizing a near **200%** increase in platform activity following in-person initiatives.
- Standardized data analysis methodologies for internal usage for long-term data tracking of future outreach effectiveness.

2D-World Interactive Game

April 2023

- Designed and implemented a 2D tile-based world engine, utilizing **object-oriented programming** to generate a real-time interactive game using **Java** data structures and packages.
- Integrated a dynamic **user interface** with unique game features, enhancing player encounters, modular level progression, and responsive data tracking.

LEADERSHIP

Ighota! | Vice President

June 2021 - July 2024

- Curated STEM curriculum for lab material and designed **15+** graphic review sheets, improving local access to education.
- Coordinated with **20+** Nigerian secondary schools, expanding STEM education resources for **600+** students.
- Led outreach efforts to several Nigerian institutions, partnering with **8+** active educators to grow student material.

Taiwanese American Student Association | Internal Vice President

December 2023 - May 2024

- Successfully executed **5+** general events for **100+** students, promoting Taiwanese culture to the campus community.
- Organized the semesterly retreat, weekly team socials and meetings, and bonding activities for **30+** cabinet members.
- Actively fostered strong camaraderie, cabinet bonding, and cross-functional teamwork among members across **7** teams.

SKILLS & INTERESTS

- Skills & Knowledge:** Python, Java, SQL, HTML, CSS, JavaScript, React.js, Git, jQuery, Pandas, Sklearn, PyTorch
- Interests:** Fullstack, AI/ML, Data Analytics & Engineering, Web Development, Product Design, Sketching, Photography